

Pattern Recognition and Machine Learning (EE602L)

Lab 5: Decision Trees for Classification and Regression

Instructions:

- Use Google Colab.
- Submit one well-commented notebook (Lab5-RollNum.ipynb) along with a PDF report (Lab5-RollNum.pdf) containing experimental outputs, analysis of the results, and conclusions.
- Wrong file names or missing files won't be graded.
- **Deadline:** 14/09/2026 11.59 PM. Late submissions lose 10% per day.
- This is an individual assignment. Please follow the Institute's Academic Code of Honor.

Question 1 — Decision Tree Classification

Problem Statement

Dataset: Iris dataset, available directly through scikit-learn (from `sklearn.datasets import load_iris`).

Train and analyze a Decision Tree classifier on the Iris dataset. Compare entropy and Gini as splitting criteria, study how tree depth leads to overfitting, and apply both pre-pruning and post-pruning to control it.

(a) Data preparation and tree training

Split the dataset into training, validation, and test sets in a 70:15:15 ratio using a fixed random seed of 42 and stratified sampling. Train Decision Tree classifiers using both entropy and Gini as the splitting criterion for at least six values of `max_depth`, including an unrestricted tree.

For each configuration, record the training and validation accuracy.

(b) Overfitting and pruning

Plot training and validation accuracy against `max_depth` for both criteria. Identify whether and approximately where overfitting begins, and select a suitable `max_depth` using validation performance.

Take the better-performing splitting criterion based on validation accuracy and select the best `max_depth` for that criterion. Then train an unrestricted tree using the same criterion and apply post-pruning using `ccp_alpha` for at least four values. Select the best `ccp_alpha` using validation accuracy and compare the resulting post-pruned model with the pre-pruned model.

(c) Evaluation and analysis

Evaluate the selected pre-pruned and post-pruned models on the test set using accuracy, macro-averaged precision, recall, and F1-score. Visualize the selected tree using an appropriate method such as `plot_tree`.

Briefly discuss the difference between entropy and Gini, why increasing tree depth can cause overfitting, and how pre-pruning and post-pruning help control it.

Question 2 — Decision Tree Regression

Problem Statement

Dataset: Diabetes dataset, available directly through scikit-learn (from `sklearn.datasets import load_diabetes`).

Train and analyze a Decision Tree Regressor on the Diabetes dataset. Understand how a regression tree selects splits and forms leaf predictions, and study how tree depth and minimum leaf size affect generalization.

(a) Data preparation and tree training

Split the dataset into training, validation, and test sets in a 70:15:15 ratio using a fixed random seed of 42. Train a `DecisionTreeRegressor` for at least six values of `max_depth`, including an unrestricted tree.

Record the training and validation MSE for each configuration.

(b) Hyperparameter study

Fix `max_depth` at a large value (e.g., 10) and vary `min_samples_leaf` across at least four values. Record the training and validation MSE for each configuration.

(c) Evaluation and analysis

Plot training and validation MSE against `max_depth` and separately against `min_samples_leaf`. Select the best model using validation MSE and evaluate it on the test set using MSE, RMSE, and MAE. Visualize the selected tree, or a shallow version of it if the full tree is too large.

Briefly explain how a regression tree uses MSE reduction to select splits, how the mean target value is used for prediction at a leaf, and how `max_depth` and `min_samples_leaf` affect overfitting and generalization.